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United States Patent Application Publication

20250277348

Kind Code

A1

Publication Date

September 04, 2025

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PILE INSTALLATION SYSTEM

Abstract

A foundation pile installation system includes a hollow foundation pile having a pair of tapered drive bar slots opening in its top end and a pair of attachment pin holes spaced below and aligned with the drive bar slots, a drive head having drive bar holes and a radially extending flange, a drive bar, and an attachment pin, the drive head for insertion in the top end of the pile in an assembled configuration in which the drive bar is inserted in the drive bar holes in the drive head and received in the drive bar slots of the pile and the attachment pin is inserted in the attachment pin holes of the drive head and in the attachment pin holes of the pile to secure the drive head to the pile, the tapered drive pin slots guiding the drive bar into the drive slots as the drive head is inserted in the pile.

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Family ID: 85178237

Appl. No.: 19/212307

Filed: May 19, 2025

Related U.S. Application Data

parent US division 17889259 20220816 parent-grant-document US 12305350 child US 19212307
us-provisional-application US 63233672 20210816

Publication Classification

Int. Cl.: E02D5/56 (20060101); E02D7/22 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Divisional application of U.S. application Ser. No. 17/889,259 filed Aug. 16, 2022, which claims the benefit of U.S. Provisional Application No. 63/233,672 filed Aug. 16, 2021, both of which are incorporated herein fully by reference.

BACKGROUND

Technical Field

[0002] The present invention relates generally to driving foundation piles into the ground and more particularly relates to a system for installing hollow foundation piles using a pile having one or more pairs of drive slots.

Description of Related Art

[0003] The traditional method for driving a hollow foundation pile in the ground uses a pile having two pairs of axially spaced drive pin holes for attachment of a drive head. In hard soils, the drive head can be attached to the pile with two pins, otherwise one drive pin is usually sufficient. When the pile has been driven the pin or pins are removed and the drive head detached. While effective, it is time consuming and cumbersome to attach and remove the pile from the drive head and limits the depth to which the pile can be driven into the ground since the drive pins must at some point be removed.

[0004] There is therefore a need for a pile installation system that facilitates attachment and detachment of the pile from the drive head and allows the pile to be driven to a desired depth without concern for removing drive or attachment pins from the pile.

SUMMARY OF THE INVENTION

[0005] A pile installation system according to the invention includes a pile having one or more pairs of cooperating drive bar slots opening upwardly on the top end of the pile and one or more pairs of attachment pin holes axially spaced below the lower ends of the pile drive bar slots. A drive head includes a drive collar having a hollow interior, a lid covering the top of the drive collar, and one or more drive bars affixed to the underside of the lid. The pile is engaged with the drive head by inserting the pile in the drive collar, rotating the pile until the drive bar slots radially align with the drive bars, and further inserting the pile in the drive collar until the drive bars engage the drive bar slots and the top of the pile abuts the lid. The pile may further be attached to the drive head by inserting an attachment pin in cooperating attachment pin holes in the drive collar and pile.

[0006] A second embodiment of a pile installation system includes a pile having a tubular body, a pair of drive pin slots and a pair of attachment pin holes axially spaced below the drive pin slots on opposite sides of the top end of the pile. A drive head includes a drive shaft and a circular flange extending radially from the drive shaft. The drive shaft includes a pair of drive pin holes and a pair of attachment pin holes axially spaced below the drive pin holes. The pile is attached to the drive head by preinstalling a drive pin in the drive pin holes in the drive shaft, inserting the drive shaft into the pile, aligning the drive pin with the drive pin slots in the pile and fully inserting the drive shaft in the pile until the drive pin engages the lower ends of the drive pin slots. An attachment pin may then be inserted into cooperating attachment pin holes in the pile the cooperate with the attachment pin holes in the drive shaft to secure the pile to the drive head.

[0007] In both embodiments, once the pile is attached to the drive mechanism, the pile can be picked up and positioned where desired for driving into the ground.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIGS. **1A** and **1B** are upper right and upper left perspective views, respectively, of a foundation pile having two pairs of drive pin slots according to the invention.

[0009] FIG. **2A** is an exploded upper perspective view of a drive head and a pile installation tool according to the invention.

[0010] FIG. **2B** is an upper perspective view of the drive head and pile installation tool shown in FIG. **2A** in an assembled configuration.

[0011] FIG. **3A** is a exploded upper perspective view of a foundation pile and a drive head connected to a pile installation tool according to the invention.

[0012] FIG. **3B** is an upper perspective view of the foundation pile and a drive head connected to a pile installation tool shown in FIG. **3A** in an assembled configuration.

[0013] FIG. **4A** is an elevational view of the drive head shown in FIGS. **2A-3B**.

[0014] FIG. **4B** is a bottom plan view of the drive head shown in FIG. **4A** taken along lines **4B-4B**.

[0015] FIG. **4C** is a sectional view of the drive head shown in FIG. **4A** taken along lines **4C-4C**.

[0016] FIG. **5** is an upper perspective view of an attachment pin used to connect the drive head to the foundation pile.

[0017] FIG. **6** is a lower perspective view of the drive head shown in FIGS. **4A** and **4B** with a cut away to show the interior of the drive head.

[0018] FIGS. **7A** and **7B** are right and left upper perspective views of a second embodiment of a pile according to the invention.

[0019] FIGS. **8A** and **8B** are exploded and assembled lower perspective views thereof.

[0020] FIG. **9** depicts plan views of drive and attachment pins thereof.

[0021] FIGS. **10A-10C** are lower perspective views of a drive head being attached to a pile according to the second embodiment.

AMENDMENTS TO THE SPECIFICATION

[0022] A foundation pile installation system according to the invention comprises a pile **10**, seen in FIGS. **1A** and **1B**, and a drive head **12** and a pile installation tool **20**, seen in FIGS. **2A** and **2B**. The pile **10** is a hollow pile, generally having a helical flange **14**, the pile being of the type that is installed in the ground using rotation and downward pressure (known in the field as “crowd”).

Whereas a prior art pile has two pairs of pin receiving holes at the top end of the pile for receiving pins with which to attach a pile installation tool to the pile, a pile, as seen in FIGS. **1A** and **1B** according to the invention, has two pairs of drive bar slots **16** on opposite sides of the pile and a single pair of pile attachment pin holes **18** as seen below the drive bar slots. In the illustrated embodiment, two pairs of drive bar slots **16** are disposed at 90° angles around the top of the pile. In another embodiment, only one pair or more than two drive bar slots are provided.

[0023] As seen in FIGS. **2A** and **2B**, the drive head **12** is configured for attachment to a pile installation tool **20** and includes a drive head cap **21**, a lid **22** and a drive collar **24** descending from the lid. The drive head cap **21** includes drive cap attachment holes **25** for receiving a drive head attachment pin **33** and the drive collar **24** has drive collar attachment pin holes **26** for receiving a pile attachment pin **34** (discussed below).

[0024] With reference now to FIGS. **3A-3B** and **4A-4C**, the hollow interior of the drive collar **24** is sized to receive a pile **10** and includes a pair of drive bars **30** arranged in a cross-key configuration (see FIGS. **4B** and **6**). The drive bars **30** are directly secured to the underside of lid **22** and are sized to be closely received in drive bar slots **16** in the top of pile **10**. While the illustrated embodiment shows a pair of drive bars, in another embodiment one or more than two drive bars may be provided.

[0025] As seen in FIGS. **3A** and **3B** the drive head **12** is attached to the pile **10** by lowering the

drive collar **24** over the top of the pile **10**, twisting the pile until the drive bar slots **16** are radially aligned with the drive bars **30**, then further lowering the drive collar **24** over the pile **10** until the drive bars **30** are receiving in the drive bar slots **16**. It will generally be necessary to pick up the pile to move it to where it will be installed in the ground, so a pile attachment pin **34** can be installed in the pile attachment pin holes **18** in the pile **10** and in the cooperating drive collar attachment pin holes **26** in the drive collar **12** to secure the pile to the drive collar to allow the pile to be picked up by the pile installation tool **20** and moved to its installation location. Once the pile is established in the ground, the pile attachment pin **34** can be removed. Downward pressure by the pile installation tool and drive head on the pile will maintain the drive bars **30** in engagement with the drive bar slots **16** on the top of the pile. Once the pile has been driven to a desired depth, the drive collar **24** can be removed without the necessity of removing any attachment pins.

[0026] In another embodiment of the invention, shown in FIGS. **7A-10C**, pile **110** includes a pair of upwardly opening drive bar slots **116** disposed on opposite sides of the top end of the pile as best seen in FIGS. **7A-7B**. The illustrated embodiment shows one pair of drive bar slots **116**, but other embodiments may include two or more pairs of drive bar slots **116**. Attachment pin holes **118** are axially spaced below the lower ends **119** of the drive pin slots **116** and radially aligned with the drive pin slots **116**. A helical flight **114** is provided towards the lower end of the pile.

[0027] In FIGS. **8A-8B** it is seen that a drive head **112** is attached by inserting a drive head attachment pin **133** into drive cap attachment holes **125** provided in a drive head cap **121** on a draft shaft **124** and into drive rig attachment holes **135** in driving rig **120**. The drive head **112** includes a circular flange **122** that extends radially from the drive shaft **124**. Drive bar holes **126** and attachment pin holes **128** are provided in the drive shaft **124** to receive drive bar **130** and attachment pin **134** (see FIGS. **9** and **10B**), respectively.

[0028] With reference now to FIGS. **10A** and **10B**, it is seen that the drive head **112** is attached to the pile **110** by preinstalling a drive bar **130** in the drive bar holes **126** in the drive shaft **124** and then inserting the drive shaft **124** into the hollow interior of the pile **110** as shown. In the illustrated embodiment, each drive bar slot **116** has tapered edges **132** to guide the drive bar **130** into the slot **116** as the drive head **112** is inserted into the pile. When the drive head **112** is fully inserted into the pile **110**, the drive bar **130** orients the pile **110** so that the attachment pin holes **128** of the drive shaft **124** align with the attachment pin holes **118** of the pile **110**, whereupon an attachment pin **134** can, if needed, be inserted mutually through the attachment pin holes **118** of the pile **110** and the attachment pin holes **128** of the drive shaft **124** to secure the drive shaft **124** to the pile **110**. The pile can then be positioned where desired and driven into the soil. When the pile **110** has been driven down to a level where the attachment pin **134** is just above the ground, it can be removed and the pile **110** driven further into the ground, using only the drive bar **130**, to a desired depth. Finally, when the pile has been driven to its specified depth, the drive shaft **124** can easily be separated from the pile simply by lifting it up. The drive slots **116** allow the pile to be driven to its specified depth even though the drive bar **130** may be below ground level.

[0029] As seen in FIGS. **8A-8B** and **10A-10C**, dual fins **138** descend from the lid **122** of the drive head **112** which together excavate an annular trench in the ground around the top of the pile **110** as it nears a desired installation depth. The trench allows use of the drive pin slot **116** as an inspection port, provides a reference point to install a concrete vibrator in the pile to ensure proper concrete consolidation, and allows air to escape while the concrete is being vibrated.

[0030] An advantage of the second embodiment of the invention is that the drive pin slots **116** in the pile **110** facilitate securing the drive shaft **124** to and disengaging it from the pile because the drive bar **130**, once preinstalled in the drive shaft **124**, can easily be slid in and out of the drive slots **116**.

[0031] There have thus been described and illustrated certain embodiments of a foundation pile installation system according to the invention. Although the present invention has been described and illustrated in detail, it should be clearly understood that the disclosure is illustrative only and is

not to be taken as limiting, the spirit and scope of the invention being limited only by the terms of the appended claims and their legal equivalents.

Claims

1. A pile installation system, the system including a hollow foundation pile suitable for installation in the ground using rotary force and downward pressure, the system comprising: the pile having a tubular body and a top end, a pair of drive bar slots on opposite sides of the body, said drive bar slots each extending axially from, and opening on, said top end, a pair of attachment pin holes axially spaced from and radially aligned with the drive bar slots, a drive head having a drive shaft including a radially extending circular flange, the drive shaft having a pair of drive bar holes and a pair of attachment pin holes, the attachment pin holes axially spaced from the drive pin holes, a drive bar, an attachment pin, and an assembled configuration in which the drive shaft is received in the pile, the top end of the pile is abutting the circular flange, the drive bar is inserted in the drive pin holes of the drive shaft and received in the drive bar slots of the pile, and the attachment pin is received in the attachment pin holes of the pile and in the attachment pin holes of the drive shaft thereby securing the drive head to the pile.
 2. The pile installation system of claim 1 further comprising: the drive bar slots in the pile each having a lower end, the attachment pin holes in the pile axially spaced from the lower ends of the drive bar slots, the drive bar holes and attachment pin holes of the drive head axially spaced apart the same distance as that between the lower ends of the drive bar slots and the attachment pin holes of the pile, such that in the assembled configuration when the drive bar is in abutting engagement with the bottom ends of the drive bar slots of the pile the attachment pin holes of the drive head axially align with the attachment pin holes of the pile.
 3. The pile installation system of claim 1 further comprising: each of the drive bar slots having a tapered opening for guiding a drive bar inserted in the drive pin holes of the drive head into the drive bar slots as the drive head is inserted into the pile.
 4. The pile installation system of claim 1 further comprising: the pile having a helical flange.
 5. A method for installing a foundation pile in the ground, the system comprising: inserting a drive bar in a pair of drive bar holes in a drive head, inserting the drive head into the hollow interior of the pile, aligning the drive bar with a pair of drive bar slots in the top end of a tubular pile, and sliding the drive shaft into the pile until the drive pin engages with the lower ends of the drive bar slots to attach the drive head to the pile.
 6. The method for installing a foundation pile of claim 5, the method further comprising: applying downward pressure and torque to the drive head thence to the pile to drive the pile into the ground.
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